

Future Improvements

1. Normalize Project Ownership

Replace owner text with owner_id FK to Users.

2. Add User Contact Information

Add email, department, team and location.

3. Standardize Lookup Values

Create lookup tables for Status, Priority, Severity, Environment and Test Type.

4. Add Audit Columns

Include created_by, updated_by, created_at and updated_at.

5. Enhance Release Tracking

Add release_manager, release_type, deployment_window, deployment_status and rollback_required.

6. Introduce Sprint Management

Create a Sprints table and link Releases, Bugs and Test Cases.

7. Implement Role-Based User Management

Create a Roles table instead of storing role names as text.

8. Improve Bug Lifecycle Tracking

Create a Bug_History table for status transitions.

9. Store Attachments Separately

Create an Attachments table linked to Bugs.

10. Optimize Performance

Add indexes on release_id, project_id, reporter_id, assignee_id and status.

Portfolio Note

This project was intentionally designed as a simplified Release Quality Analytics Platform to demonstrate relational database modeling, SQL querying, reporting techniques, and data analysis concepts. The listed improvements represent potential enhancements for scaling the solution toward an enterprise-level software quality management system.